

Project Design Phase-II

Technology Stack

Date	30 September 2026	Team ID	SWTID-2026-1617
Project Name	Import Data using Transform Maps (Spreadsheet)	Maximum Marks	4 Marks

Technical Architecture:

The architecture describes how spreadsheet data is prepared, imported into ServiceNow using System Import Sets and Transform Maps, mapped to the Employee Training Records table, and secured using users, roles and ACLs. Employee reference data can be accessed through dot-walking for related information such as Department.



Architecture Flow

Spreadsheet → System Import Sets → Transform Map & Field Maps → Employee Training Records → Employee Dot-Walking → Roles & ACLs → Access Validation

Technology focus: ServiceNow Platform UI, System Import Sets, Transform Maps, Field Maps, ServiceNow Tables, User/Role Management, ACL Engine, ServiceNow platform storage and JavaScript-based configuration where required. No external API or machine-learning model is required for this workflow.

Table-1: Components & Technologies

Technology components required to implement the spreadsheet import, data relationship and security workflow.

S.No	Component	Description	Technology
1	User Interface	Administrators and developers prepare data, configure the ServiceNow instance and verify imported records and access.	ServiceNow Platform UI / Web UI
2	Data Source	Spreadsheet contains the four required source fields: Training Name, Completion Date, Status and Employee.	Spreadsheet / XLSX
3	Import Layer	Loads the spreadsheet into ServiceNow and creates the import source for Employee Training data.	System Import Sets / Load Data
4	Transformation Layer	Maps source spreadsheet fields to target table fields and executes the data transformation.	Transform Maps / Field Maps
5	Application Table	Stores imported Employee Training records and their related employee information.	ServiceNow Custom Table / Employee Training Records
6	Related Data	Retrieves related Employee information such as Department from the referenced Employee record.	Reference Fields / Dot-Walking
7	User & Role Management	Creates test users and assigns appropriate roles for controlled access to Employee Training Records.	ServiceNow Users / Roles / Groups
8	Security Layer	Controls Read and Write access to Employee Training Records based on configured permissions.	ServiceNow ACL Engine / Access Control Lists
9	Platform Storage	Stores tables, imported records, user/role configuration, mappings and security configuration.	ServiceNow Platform Database / Configuration Storage
10	External API	No external API is required for the spreadsheet import and security workflow.	Not Applicable
11	Machine Learning Model	No machine-learning model is required because the workflow is configuration- and rule-based.	Not Applicable
12	Infrastructure	Hosts the ServiceNow application, data import process, configuration and security controls.	ServiceNow Cloud Instance

Table-2: Application Characteristics

Key characteristics of the proposed ServiceNow import and security solution.

S.No	Characteristics	Description	Technology
1	Data Integration	Imports employee training data from a spreadsheet and maps source fields to the ServiceNow target table.	System Import Sets / Transform Maps
2	Security Implementations	Protects imported records through user roles and ACLs. Read and Write access is configured and validated.	ServiceNow ACLs / Roles / Groups
3	Data Relationships	Links each training record to an Employee and uses dot-walking to access related Department information.	Reference Fields / Dot-Walking
4	Scalable Architecture	Centralized import mappings and role-based security can be reused as additional training records and users are added.	Transform Maps / ServiceNow Platform
5	Availability	The workflow is available whenever the ServiceNow instance and required platform features are available.	ServiceNow Cloud Platform
6	Performance	Use clear field mappings and efficient access rules to keep import and record-access operations responsive.	Transform Maps / ACL Engine
7	Reliability	Consistent source-to-target mapping and verification help ensure accurate imported records.	Field Maps / Transform / Testing
8	Maintainability	Import mappings, roles and ACL configuration can be reviewed and updated when requirements change.	ServiceNow Configuration
9	Testing & Validation	Impersonate authorized and unauthorized users, search Employee Training Records and verify expected access.	ServiceNow Impersonation / ACL Testing
10	External Dependencies	The core workflow does not depend on external APIs or machine-learning services.	Not Applicable

Technology Summary: The project uses ServiceNow Platform UI, Spreadsheet/XLSX, System Import Sets, Transform Maps, Field Maps, ServiceNow Tables, Reference Fields/Dot-Walking, Users/Roles/Groups and ACLs. The workflow is tested through user impersonation and access verification. No external API or machine-learning model is required.